Nathan Houck

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EXECUTIVE SUMMARY

Aerospace engineering student at Embry-Riddle Aeronautical University specializing in astronautics, with hands-on experience designing, prototyping, and testing mechanical and thermal systems for real-world aerospace applications. Demonstrated ability to lead technical teams and execute engineering projects through full development cycles.. Seeking a Summer 2026 test engineering internship at Northrop Grumman to contribute analytical rigor and practical engineering skills to defense systems testing.

EDUCATION

Embry-Riddle Aeronautical University (ERAU) Daytona Beach, FL
Bachelor of Science, Aerospace Engineering May 2027
Area of Concentration: Astronautics | GPA: 3.75/4.0

- Dean's List, Fall 2023, Spring 2024, Spring 2025, Fall 2025
- Honor Roll, Fall 2024
- Exceptional Eagles Award – ERAU Student Involvement Awards, Spring 2025
- Presidential Scholarship, Fall 2023 – Present

Bayard Rustin High School West Chester, PA
High School Diploma June 2023
GPA: 4.4/4.0

PROJECT EXPERIENCE

Autonomous Maritime Robotics Association – Robosub Aug. 2024 – Aug. 2025
Heat Transfer Team Lead

- Researched a solution for electronics cooling within the submarine to reduce internal temperature by over 60°F.
- Investigated heat pipes and thermoelectric cooling.

Mechanical Engineering Team

- Worked with a team of 4 to design, prototype, and manufacture an improved thruster mount.
- Reduced lateral movement to increase reliability and lifespan.
- Designed and modeled a torpedo launching system that launches the torpedoes when the proper target is identified.
- Prototyped a bin that will release weighted bags over a designated drop zone, utilizing torsion springs and servo motors.
  - Created an IP68 compliant servo housing for a DS3225 servo motor utilizing O-Rings and a dynamic shaft seal.

Experimental Jet Engine Propulsion Club (XJEP) Aug. 2025 – Present
Aerospace Engineer

- Designing a novel cooling solution for a modular afterburner attached to a KingTech turbine engine.
- Working on a paper to be published in AIAA's SciTech convention studying afterburner efficiency.

RESEARCH

Undergraduate Researcher – Thermal Science Laboratory (ERAU) Aug. 2025 – Dec. 2025

- Collaborated with the Instituto Tecnológico de Aeronáutica to create a flow visualization model to study the behavior of phase change materials.
- Created a test section using Autodesk Inventor to contain the PCM and allow it to be oriented at any angle.
  - The test section simulated a constant heat flux boundary condition and a constant temperature boundary condition.

Personal Research – Houck Airfoil Patent US9976421B2 Sep. 2025 – Present

- Researching the patented lifting foil design and identifying potential, practical applications for aerospace technologies.
- Evaluating the technology readiness level for further independent study with the intent to commercialize the technology.

WORK EXPERIENCE

Embry-Riddle Aeronautical University | Grading Assistant Jan. 2026 – Present

- Grading and evaluating homework assignments for AE319 Aerodynamics, ensuring accuracy and adherence to course standards.

- Providing written feedback to support student understanding.
- Maintaining organized grading records and meeting strict deadlines outlined by the course instructor.

Wawa, Inc. | Sales Associate 2022 – Present

- Worked in a fast-paced environment on a strict schedule to serve over 500 customers in one shift.
- Handled day-to-day operation of the store, including the sales floor, deli area, and the register.
- Trained new sales associates in company policies and customer service techniques.
- Demonstrated a high level of customer service to every customer; received Day Brightener Award.

Greater Philadelphia YMCA | Lifeguard 2022 – 2023

- Maintained safety standards in and around the pool to prevent injury.
- Ensured safety and quality of pool water by testing chemical levels.
- Maintained CPR/AED and First Aid certification.

LEADERSHIP / INVOLVEMENT

Undergraduate Researcher, Thermal Science Laboratory, Fall 2025
Member, Experimental Jet Engine Propulsion Club, Fall 2025 – Present
Member, AIAA, Fall 2025 – Present
Flight Student, Summer 2024 – Present

- Risk management and avoidance
- Effective communication and situational awareness
- Problem solving and remaining calm under pressure
- 28.1 Flight Hours (not current)

Member, Diving Eagles Scuba Club
Member, National German Honor Society, Fall 2022 – Spring 2023
Volunteer, Event Supervisor, Science Olympiad Regionals, Fall 2019 – Present

SKILLS

Engineering Software: CATIA v.5, Autodesk Fusion, Autodesk Inventor, MATLAB, Python, Java, C/C++ (Basic), ANSYS Fluent (Basic)
Office Software: Microsoft Word, Excel, PowerPoint, Publisher, Teams
Technical: Drafting, 3-D Printing, Computer-Aided Design, CPR/AED and First Aid Certified
Languages: English (fluent), German (basic)
Hobbies and Interests: Aviation, Guitar, Alto Saxophone, Scuba Diving, Fossils